The reasons for the growth and decline of a country's population

The **birth rate** is defined as the number of births per thousand population in a year. If the birth rate of a country is 17/1000 (17 per 1000), this means that, on average, for every 1000 people in this country, 17 births will occur in a year. The **death rate** is the number of deaths per thousand population in a year. If the death rate for the same country is 8/1000, it means that, on average, for every 1000 people, eight deaths will occur in a year.

The difference between the birth rate and the death rate is called the **rate of natural change**. If it is positive, it is termed natural increase. If it is negative, it is known as natural decrease. In the case given above, there is a natural increase of 9/1000 (17/1000 – 8/1000). The rate of natural increase may also be shown as a percentage, so in this example 9/1000 is equivalent to 0.9 per cent. This is the current rate (2023) of natural increase for the world as a whole – look at the birth and death rates given in Table 6.4, which shows how much birth and death rates vary by world region.

▼ **Table 6.4** Birth and death rates by world region, 2023

Region	Population (millions)	Birth rate (per 1000)	Death rate (per 1000)
World	8009	17	8
Africa	1453	32	8
Asia	4739	14	7
Latin America/Caribbean	652	14	8
North America	375	11	10
Europe	744	9	12
Oceania	45	15	7

Natural change and net migration

Population change in a country is affected by:

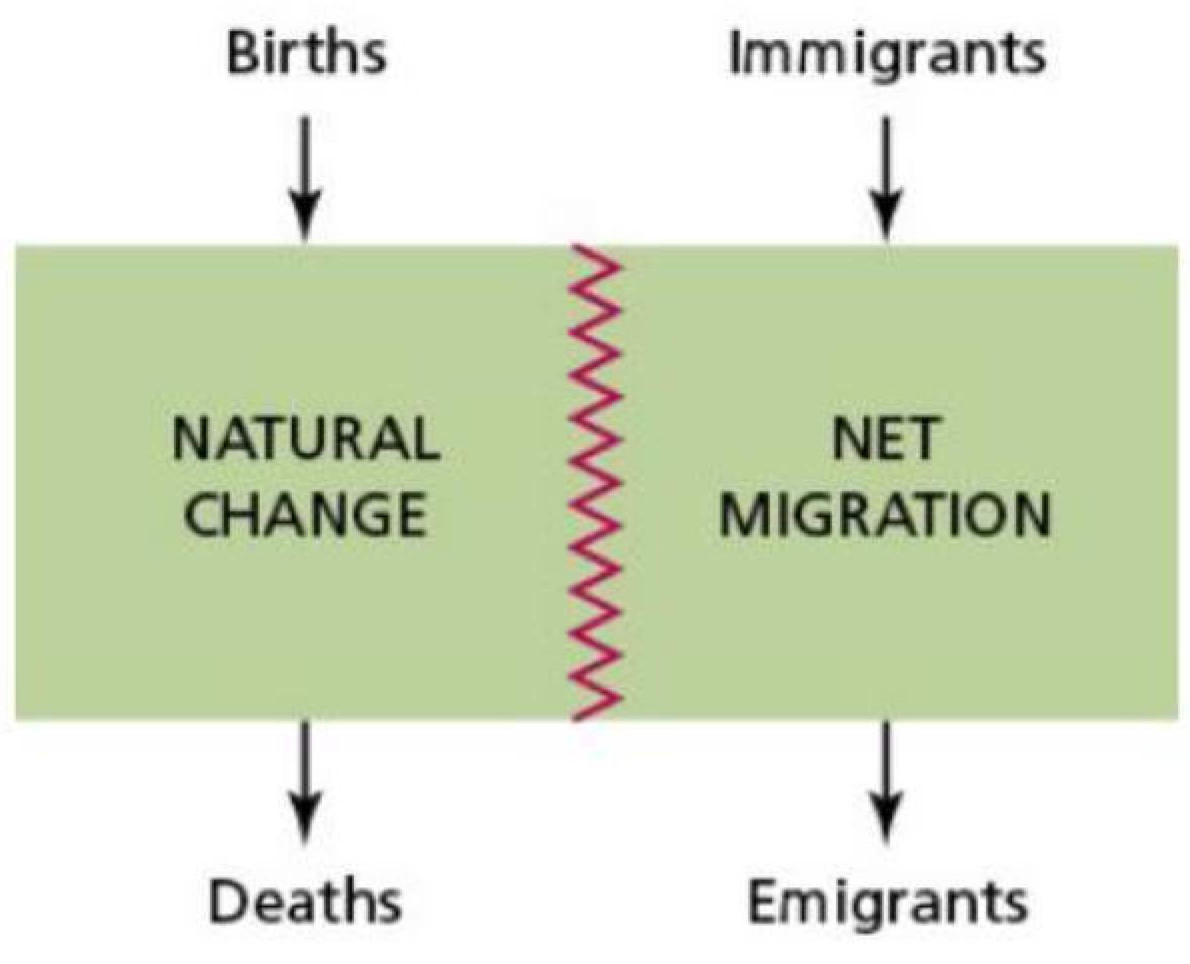
» natural change – the difference between births and deaths
» **net migration** – the balance between immigration and emigration.

The **immigration rate** is the number of immigrants per 1000 population entering a receiving country in a year. The **emigration rate** is the number of emigrants per 1000 population leaving a country of origin in a year.

The dividing zigzag red line on Figure 6.5 indicates that the relative contributions of natural change and net migration can vary over time within a particular country, as well as between countries at any one point in time. The model is a simple graphical alternative to the popular equation:

$$P = (B - D) +/- M$$

Where P = population, B = births, D = deaths, and M = migration



▲ **Figure 6.5** Input–output model of population change

Figure 6.6 shows some simple demographic calculations for the imaginary island of Pacifica.

Population at beginning of year: 5000

Population change during the year:
Births: 150 Deaths: 60
Immigrants: 20 Emigrants: 10

Rates of change based on data above
Birth rate: 30/1000 Death rate: 12/1000
Rate of natural change: +18/1000
Immigration rate: 4/1000 Emigration rate: 2/1000
Rate of net migration: +2/1000

Total population at end of the year
= 5100 (natural change of 90 + 10 for net migration)

▲ **Figure 6.6** Demographic calculations for the island of Pacifica

For most countries natural change is a more important factor in population change than net migration, but there are exceptions:

» Since the 1990s, positive net migration has been the primary source of population growth in HICs.
» Migration is projected to be the only driver of population growth in HICs after 2020.
» By 2050 it is expected that the population of HICs will begin to decline, as net migration will no longer be sufficient to compensate for the excess of deaths over births.

Sixty per cent of the increase in the UK's population between 2001 and 2020 was due to the direct contribution of net migration.

Activities

1 Define:
  a The birth rate
  b The death rate
  c The rate of natural change.
2 What is net migration?
3 Look at Table 6.4 (page 127). Calculate the rate of natural change for each region.
4 Look at Figure 6.6. Imagine that the population at the beginning of the year was 4000 rather than 5000. Calculate the rates of change for this new starting population.

Fertility

Definitions and global variations

The birth rate and the death rate are the most basic measures of fertility and mortality. Because these measures are so basic, demographers refer to them as the 'crude birth rate' and the 'crude death rate' because they take no account of gender or age.

Fertility varies widely around the world. It can also change significantly within a country or world region over time. For more accurate measures of fertility than the birth rate, the **fertility rate** and the **total fertility rate** are used. The crucial factor in fertility is the percentage of young women of reproductive age. Another important measure is **replacement level fertility**.

» The fertility rate is the number of live births per 1000 women aged 15–44 years in a given year.
» The total fertility rate (TFR) is the average number of children born to a woman in a country during their lifetime.
» Replacement level fertility is the level at which each generation has just enough children to replace themselves in a population. A TFR of 2.12 is generally considered to be replacement level fertility.

Table 6.5 shows how much the total fertility rate varies by world region. In terms of individual countries, the total fertility rate (mid-2023) varied from a high of 6.7 in Niger and 6.4 in the Central African Republic, to a low of 0.8 in South Korea and 0.7 in China, Macau SAR and Hong Kong SAR.

▼ **Table 6.5** The total fertility rate by world region, 2023

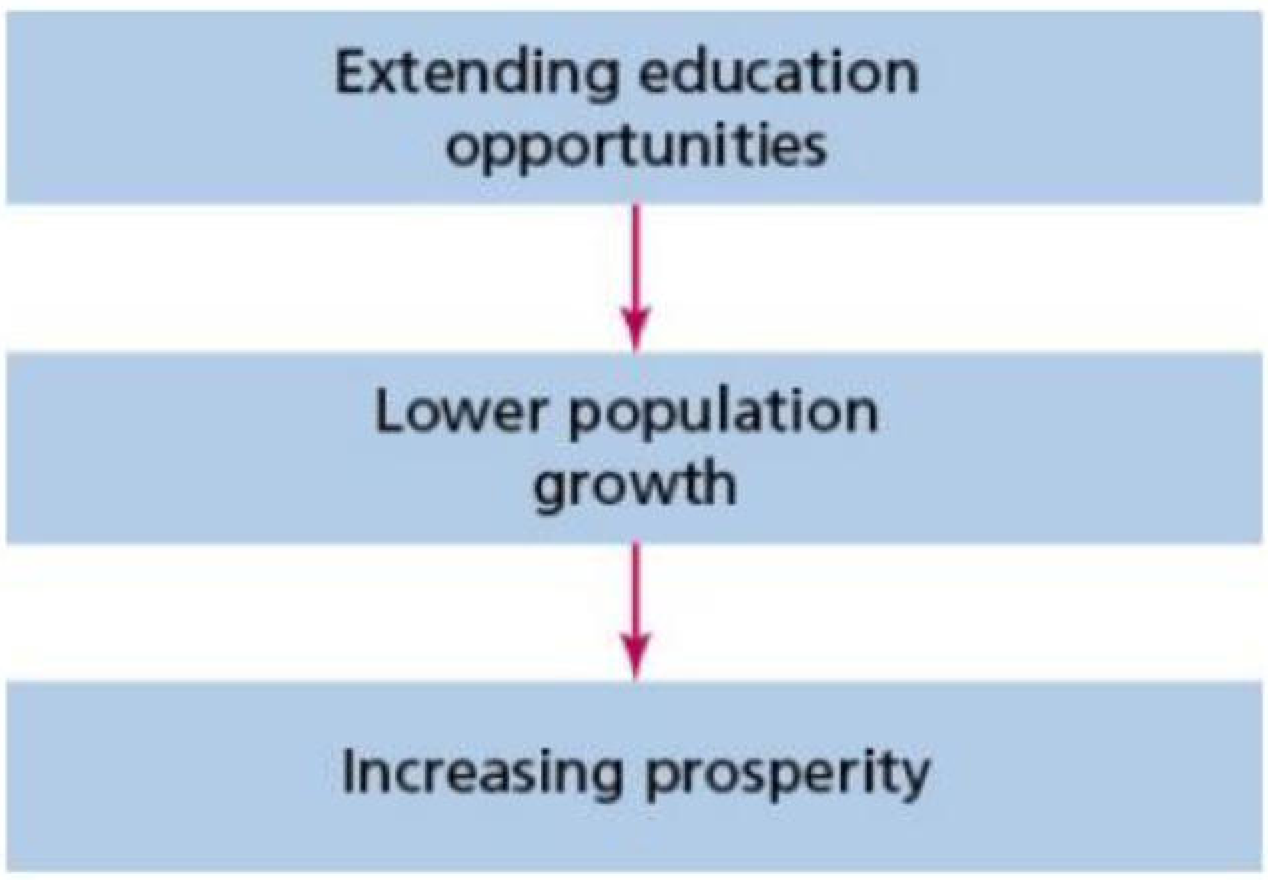
Region	Total fertility rate
World	2.2

Africa	4.3
Asia	1.9
Latin America/Caribbean	1.9
North America	1.6
Oceania	2.1
Europe	1.4

The factors affecting levels of fertility

The factors affecting fertility can be grouped into four main categories (Table 6.6).

- The most important factor is the number/proportion of women in a population in the reproduction age range, particularly in the younger part of this range.
- Education can be a major influence on fertility (Figures 6.7 and 6.8). Education can provide increased knowledge of contraception and family planning, greater social awareness, more opportunity for employment and a wider choice of action generally. There is a very strong correlation between the increasing proportion of girls enrolled in secondary education and falling rates of fertility (Figure 6.9).
- Lower population growth is genrally associated with higher economic growth. Economic growth allows greater spending on health, housing, nutrition and education, which is important in lowering mortality and in turn reducing fertility.



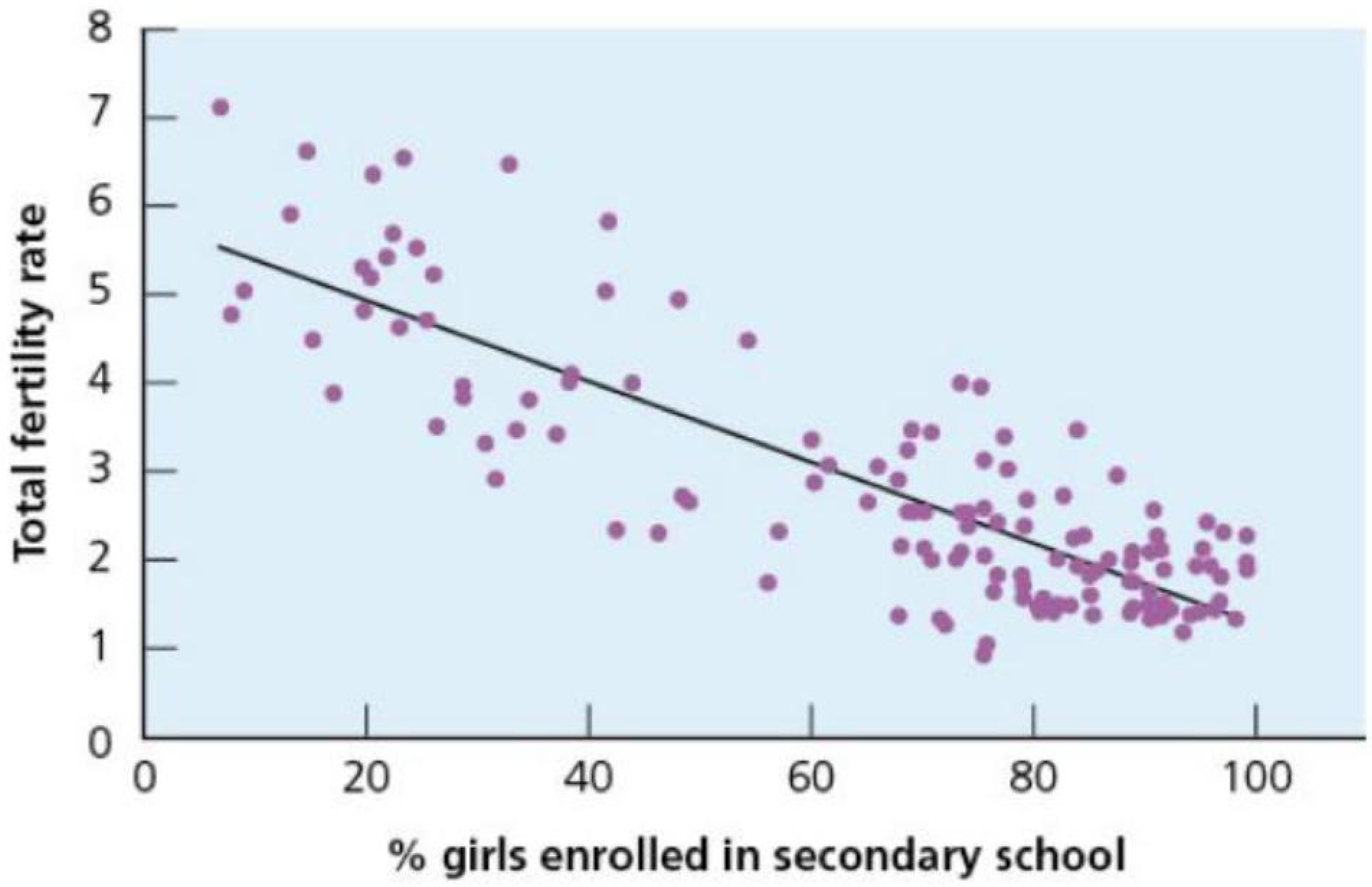
▲ **Figure 6.7** Education and development

▲ **Figure 6.8** A school in Indonesia – this country sees education as essential for its future development

▼ **Table 6.6** The factors affecting levels of fertility

Demographic	● Female age is the most important factor affecting fertility. ● When infant and child mortality rates are high, parents often have more children to compensate for these expected deaths. ● The average age of marriage impacts levels of fertility (Figure 6.10).
Social/cultural	● In some societies tradition demands high rates of reproduction. ● Education, especially female literacy, is a key factor in lowering fertility. ● In some countries religion is an important, although a generally declining, influence on fertility in terms of being comfortable with artificial contraception.

	● In most countries, fertility rates are higher in rural areas than in urban areas. ● In countries/regions where extended family networks are close by, grandparents are much more likely to help with childcare. ● In most countries, population policies have aimed to reduce fertility by investing in birth control programmes. ● Lifestyle factors (smoking, drinking alcohol, obesity, etc.) can influence reproductive health.
Economic	● In rural areas of some low-income countries, children are seen as an economic asset because of the work they do and also because of the support they are expected to give their parents in old age. ● In high-income countries the general perception is reversed, with the cost of the child-dependency years a major factor in the decision to begin or extend a family. ● The lack of affordable housing and other 'cost of living' issues have been cited as a major reason for low fertility rates in recent years in a number of countries.
Political	● At various times in the past individual countries have encouraged larger families, regarding a higher population as important for national status and power. ● The duration of internal and international conflicts can also have a huge impact on fertility.



▲ **Figure 6.9** An international comparison between female secondary school enrolment and total fertility rates

▲ **Figure 6.10** A wedding party in Tokyo, Japan – the Japanese government is trying to encourage higher fertility

Fertility decline

The global peak population has been continually revised downwards in recent decades. This is in sharp contrast to warnings in the 1960s and 1970s of a 'population explosion'. The main reason for this slowdown in population growth is that fertility levels in all parts of the world have fallen faster than was previously expected. In the second half of the 1960s, after a quarter-century of increasing growth, the rate of population growth began to slow down. Since then, many LICs and MICs have seen the quickest falls in fertility ever known and thus the earlier population projections did not materialise. Between 1970 and 2020, fertility declined in every country in the world:

» Globally, fertility has decreased from an average of five births per woman in 1950 to 2.2 births per women in 2023.
» Global fertility is projected to fall to 2.1 births per woman by 2050.
» With global fertility decline, a growing number of countries have reached or fallen below replacement

level fertility. In 2023, 103 countries and territories had total fertility rates below 2.1.

The movement to below replacement level fertility is undoubtedly one of the most dramatic social changes in history.

Activities

1. Define the total fertility rate.
2. Describe the differences in the total fertility rate by world region.
3. Study Table 6.6 (page 129). Discuss the influence on fertility of one factor from each of the four categories.
4. What is the evidence that global fertility is in decline?

The impacts of pro-natalist and anti-natalist policies on birth rates

In countries where governments have developed policies relating to fertility, these policies may be:

- **pro-natalist** – encouraging higher fertility
- **anti-natalist** – encouraging lower fertility.

Since the establishment of modern population policies from the 1950s, most countries that have tried to control fertility have sought to lower it. It began in earnest with India in the early 1950s. Because its family planning programme was perceived to be working, it was not long before many other developing countries followed India's policy of government investment to reduce fertility. The most extreme anti-natalist policy ever introduced commenced in China in 1979. This was known as the one-child policy.

As fertility has fallen rapidly in the vast majority of countries, it may seem obvious that family planning policies have been successful. However, a counter-argument is that, as countries develop with improvements in female literacy and other development indicators, fertility decline is at least partly due to the personal choices made by large numbers of women. With economic development, the 'real cost' of raising children becomes more obvious and the benefits of directing income in other directions become more attractive for many people. A reasonable assertion is that both factors have proceeded hand in hand and that the government policies have speeded up a process that would have taken maybe decades more to reach where it is now.

With the continuous fall in global fertility, in recent decades an increasing number of countries have sought to slow or halt fertility decline. Their concerns have centred round:

- the socioeconomic implications of their ageing populations
- the long-term prospect of absolute population decline.

Pro-natalist countries include Japan, Canada, South Korea, Singapore and many countries in Europe. France's relatively high fertility in European terms can be partly explained by its long-term active family policy, adapted in the 1980s to accommodate the entry of women into the labour force. France has taken steps to encourage fertility on a number of occasions over the last 70 years. Measures to encourage couples to have two or more children include:

- longer maternity and paternity leave
- higher child benefits
- preferential treatment in the allocation of government housing.

Overall, France has tried to reduce the economic cost to parents of having children. French economists argue that although higher fertility means more expenditure on child-care facilities and education, in the longer term it gives the country a more sustainable age structure.

In recent decades, the Chinese government has become increasingly concerned about the adverse effects of its one-child policy. These include:

- demographic ageing
- an unbalanced sex ratio
- a generation of 'spoiled' only children
- a social divide as an increasing number of wealthy people 'bought their way round' the legislation.

With growing concern about these issues, the Chinese government relaxed the rules from 2016 to allow couples to have two children. In 2021, China again amended its Population and Family Planning Law to permit couples to have three children. However, the recent efforts by the Chinese government to encourage higher fertility seem to have failed. The birth rate, which was 12.2/1000 in 2016, has declined year-on-year, to 10.9/1000 in 2023. This seems to be due to a combination of two main factors:

- New established social norms initiated by the one-child policy.
- The conditions that many Chinese people of reproductive age see as being adverse to starting or extending a family, such as the perceived high cost of raising children in cities.

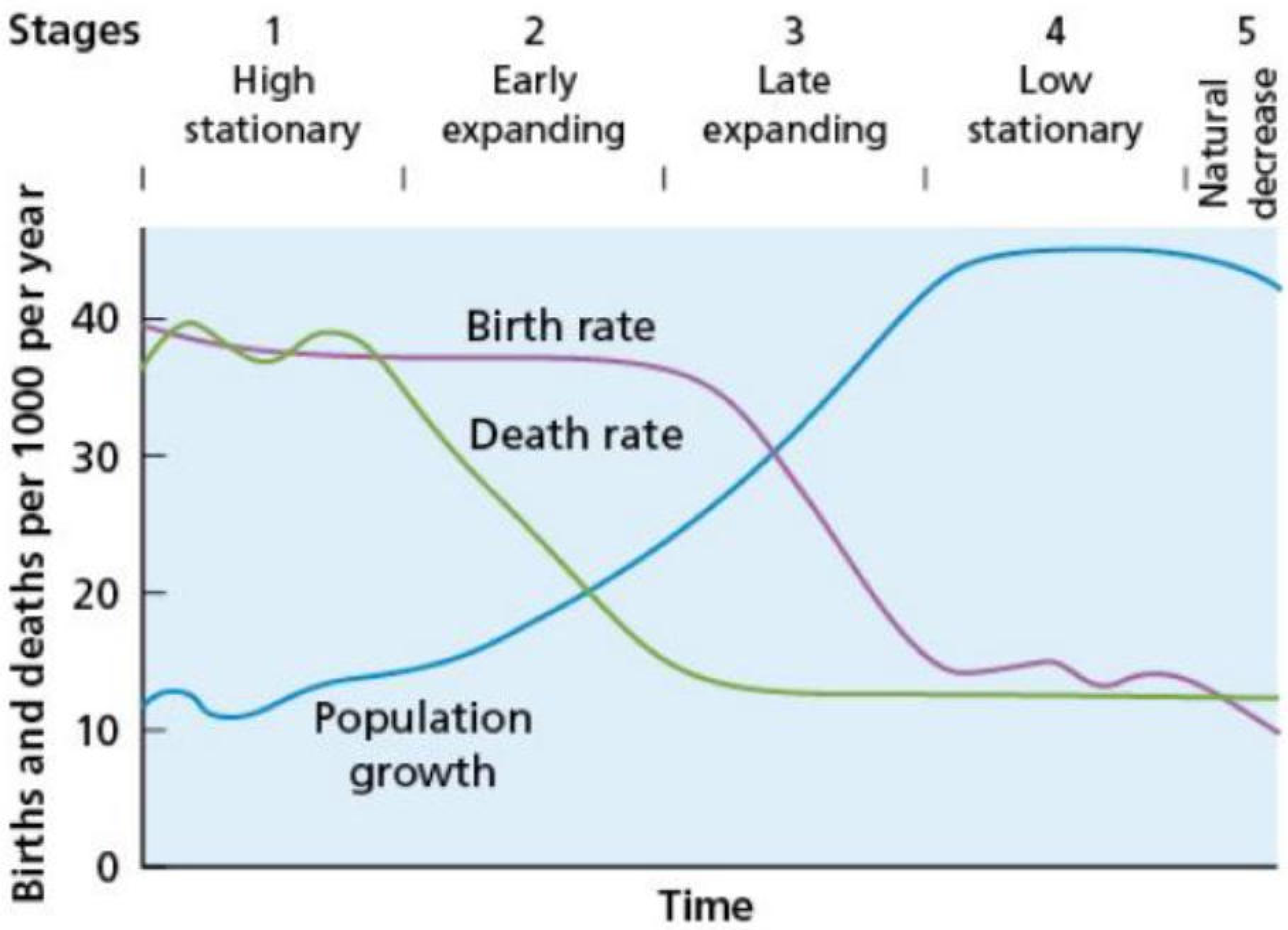
Attempting to reverse declining fertility is not easy and requires careful planning and well-targeted expenditure. A number of studies into the issue of low fertility have stressed the importance of removing the obstacles to marriage and facilitating access to decent housing. The financial concerns of young people have been an important factor in the rising average age at marriage and the increasing age of parents at the birth of first and subsequent children.

Activities

1 Explain the difference between pro-natalist and anti-natalist policies.
2 How has France tried to encourage a higher birth rate?
3 Why did China change from an anti-natalist policy to a pro-natalist policy?

The demographic transition model

Demographic transition is the change from a mainly rural, agricultural society with high fertility and mortality rates, to a largely urbanised society with much lower fertility and mortality rates. The **demographic transition model** (Figure 6.11) helps explain how and why the characteristics of a country's population change over time. A model is a simplification of reality, but it helps us understand the most important aspects of a process.



▲ **Figure 6.11** The demographic transition model

» The high stationary stage (stage 1): The birth rate is high and stable, while the death rate is high and fluctuating due to the sporadic incidence of famine, disease and war. Population growth is very slow and there may be periods of considerable decline. Infant mortality is high and life expectancy is low (Figure 6.12). A high proportion of the population is under the age of 15. Society is pre-industrial, with most people living in rural areas and dependent on subsistence agriculture.

▲ **Figure 6.12** A graveyard dating from the eighteenth century in the UK – inscriptions show that life expectancy at that time was very low

» The early expanding stage (stage 2): The death rate declines significantly. The birth rate remains at its previous level as **social norms** governing fertility take time to change. As the gap between the birth and death rates widens, the rate of natural change increases to a peak at the end of this stage. Infant mortality falls and life expectancy increases. The proportion of the population under 15 increases. The decline in mortality is mainly due to better nutrition, improved public health (particularly in terms of clean water supply and efficient sewage systems) and medical advance. Considerable rural-to-urban migration occurs during this stage.

» The late expanding stage (stage 3): After a period of time, social norms adjust to the lower level of mortality and the birth rate begins to decline. The birth rate declines due to:
  - better education about, and improved access to, contraception
  - declining infant mortality due to advances in healthcare, nutrition and public health
  - increasing gender equality and employment opportunities for women
  - changing cultural expectations about family size.

  Urbanisation generally slows and average age increases. **Life expectancy at birth** continues to increase and infant mortality to decrease. Countries in this stage usually experience lower death rates than nations in the final stage due to their relatively young population structures (Figure 6.13).

▲ **Figure 6.13** Children helping their parents milk goats in the Gobi region of southern Mongolia

» The low stationary stage (stage 4): Both birth and death rates are low. The former is generally slightly higher, fluctuating somewhat due to changing economic conditions. Population growth is slow. Death rates rise slightly as the average age of the population increases. However, life expectancy still improves as age-specific mortality rates continue to fall.

» The natural decrease stage (stage 5): In an increasing number of countries the birth rate has fallen below the death rate, which results in natural decrease. In the absence of net migration inflows, these populations decline.

No country as a whole retains the characteristics of stage 1. This stage only applies to the most remote societies on Earth, such as isolated tribes in New Guinea and the Amazon basin who have little or no contact with the outside world. LICs tend to be in stage 2, with some moving into stage 3. Most MICs are in stage 3, with some in stage 4. All HICs are now in stages 4 or 5. Stage 5, natural decrease, is mainly confined to Eastern and Southern Europe and parts of East Asia. Examples include Italy, Germany, Bulgaria, Belarus, Ukraine, Japan and South Korea.

Criticisms of the demographic transition model

Critics of the model see it as too Eurocentric, as it was based on the experience of Western Europe. They argue that many developing countries may not follow the sequence set out in the model. It has also been criticised for its failure to take into account changes due to migration.

The model presumes that all countries will eventually pass through all stages of the transition, just as most HICs have done. Because these countries have achieved economic success and enjoy generally high standards of living, completion of the demographic transition has come to be associated with socioeconomic progress, which many LICs in particular might struggle to achieve.

There are a number of important differences in the way that developing countries have undergone population change compared to the experiences of most developed countries before them:

» In developing countries, birth rates in stages 1 and 2 were generally higher than they were in developed countries when they were in these early stages.

» The death rate in developing countries fell much more steeply and for different reasons. For example, the rapid introduction of modern medicine, particularly in the form of inoculation against major diseases, has had a huge impact on lowering mortality.

» Some developing countries had much larger base populations, thus the impact of high growth in stage 2 and the early part of stage 3 has been far greater in terms of absolute population growth.

» For those developing countries in stage 3, the fall in fertility has also been steeper. This has been due mainly to the relatively widespread availability of reliable modern contraception.

» The relationship between population change and economic development has been much more tenuous in some developing countries.

Activities

1 What is a geographical model (such as the demographic transition model)?
2 Explain the reasons for falling mortality in stage 2 of demographic transition.
3 Why does it take some time before the fall in fertility follows the fall in mortality?
4 What are the characteristics of stage 5?
5 Discuss the merits and limitations of the demographic transition model.

6.2 Population structures change over time

This chapter will explain:

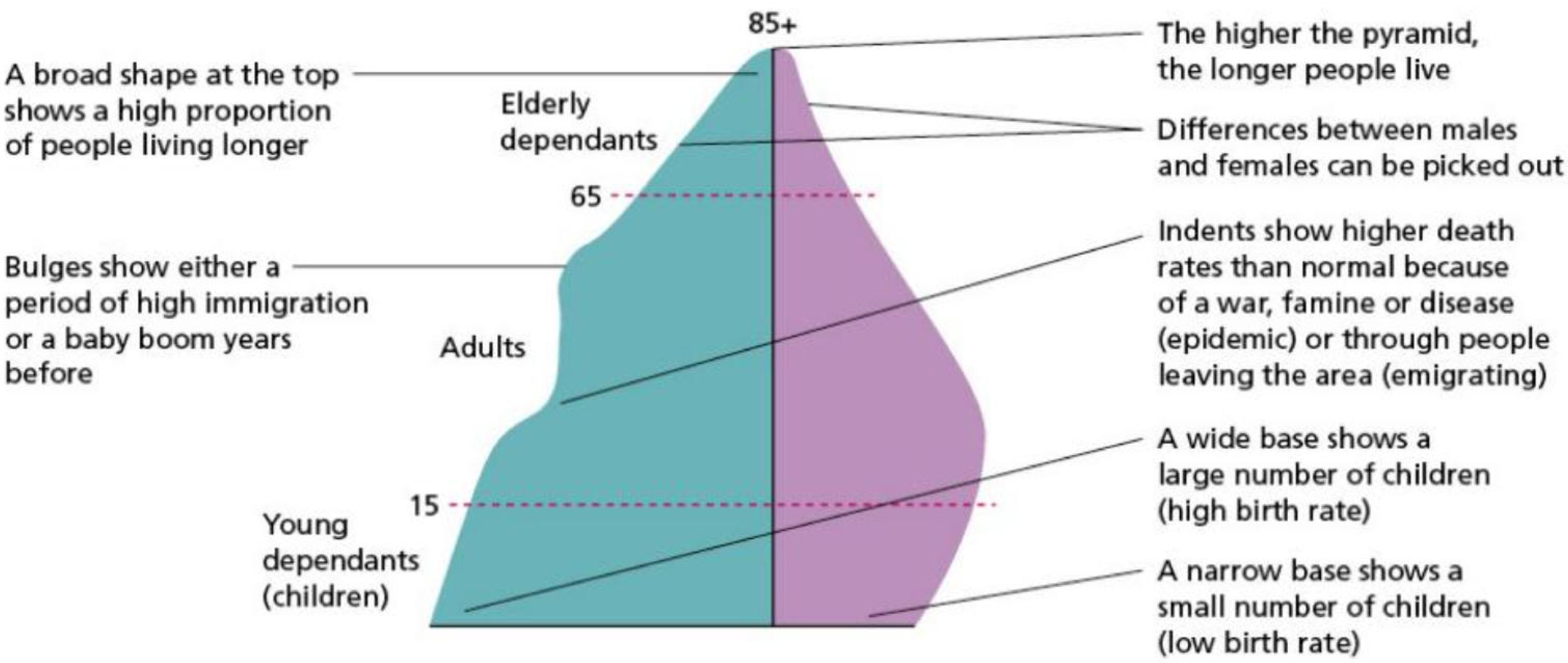
★ the factors influencing population structures
★ the issues of youthful populations
★ the issues of demographic ageing
★ detailed specific example: renewed concerns about population size in India in the twenty-first century.

The factors influencing population structures

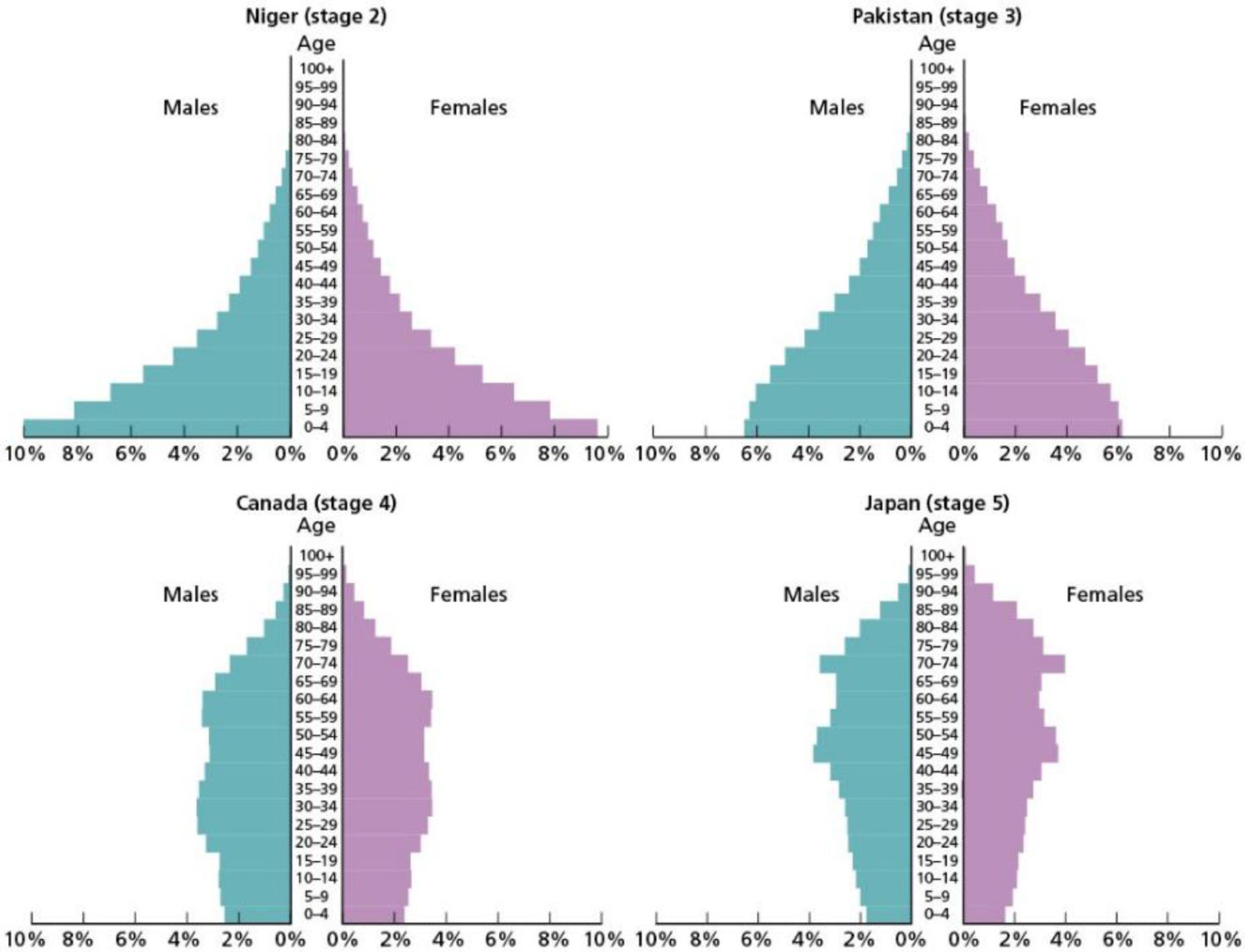
The most studied aspects of **population structure** are age and sex. Other aspects of population structure that can also be studied include race, language, religion and social/occupational group. The two fundamental influences on population structure are natural increase and net migration. These factors have already been examined in detail in Section 6.1 (page 127).

Age and sex structure are usually illustrated by the use of **population pyramids** (Figure 6.16). Pyramids can be used to show either absolute or relative data. Absolute data shows the figures in thousands or millions, while relative data shows the numbers in percentages. The latter is most frequently used as it allows for easier comparison of countries with different population sizes. Each bar represents a five-year age group. The male population is represented to the left of the vertical axis, and females to the right. Figure 6.15 provides some useful tips for understanding population pyramids.

▲ **Figure 6.14** Elderly people are making up an increasing proportion of the population in many countries



▲ **Figure 6.15** An annotated population pyramid



▲ **Figure 6.16** Four population pyramids, 2022

Population pyramids change significantly in shape as a country progresses through demographic transition. The different timing and pace of fertility decline around the world has produced distinct contrasts in age profiles. The four pyramids in Figure 6.16 relate to the demographic transition model, with Niger in stage 2, Pakistan in stage 3, Canada in stage 4 and Japan in stage 5.

» Niger (stage 2): The wide base reflects one of the highest birth rates in the world. The marked decrease in width of each successive bar indicates relatively high mortality and limited life expectancy. Youth dependency is extremely high, but aged dependency is very low.
» Pakistan (stage 3): The base of this pyramid is narrower than that of Niger, reflecting a considerable fall in the birth rate after decades of government-promoted birth control programmes. The percentage population under 15 years of age is significantly lower than that of Niger. The share of the population aged 65 and over is greater, indicating a higher life expectancy compared to Niger.
» Canada (stage 4): A lower birth rate compared to Pakistan is illustrated by a much narrower base to the pyramid. The greater uniformity of the pyramid with increasing age indicates a further decline in mortality and greater life expectancy compared to Pakistan. Aged dependency in Canada is considerably higher than in Pakistan and Niger.
» Japan (stage 5): The distinctly inverted base reflects the lowest birth rate of all four countries. The width of the rest of the pyramid is a consequence of the lowest mortality and highest life expectancy of all four countries. The birth rate is lower than the death rate.

▼ **Table 6.7** GNI per capita and demographic data for Niger, Pakistan, Canada and Japan

Indicator	**Niger**	**Pakistan**	**Canada**	**Japan**
Birth rate	45 per 1000	28 per 1000	10 per 1000	7 per 1000
Death rate	8 per 1000	7 per 1000	8 per 1000	12 per 1000
Natural change	3.7%	2.1%	0.2%	-0.5%
Infant mortality rate	40 per 1000	52 per 1000	4.5 per 1000	1.7 per 1000
Life expectancy	62	66	82	85
Population <15	49%	37%	16%	12%

Population 65+	2%	4%	19%	29%
GNI per capita (PPP)	$1330	$5800	$51,690	$44,570
Urban population	17%	37%	82%	92%

Table 6.7 compares these four countries for a range of important indicators. The difference in GNI per capita between Niger and Pakistan is consistent with the general difference between countries in stages 2 and 3 of demographic transition, although many countries in stage 3 have a higher GNI per capita than Pakistan. The much higher GNI figures for Canada and Japan are also to be expected. There is no significant difference in income per capita between countries in stages 4 and 5 of demographic transition.

The **dependency ratio** is an important measure used in the analysis of population pyramids. Dependants are viewed as people who are too young or too old to work. The dependency ratio is the relationship between the working or economically active population and the non-working population. The formula for calculating the dependency ratio is:

$$\text{Dependency ratio} = \frac{\%\text{ population aged 0–14} + \%\text{ population aged 65 and over}}{\%\text{ population aged 15–64}} (\times 100)$$

A dependency ratio of 80 means that for every 100 people in the economically active population, there are 80 people dependent on them. In this example, the dependency ratio is related to the base number of 100. However, sometimes it is expressed in relation to the base number of 1. In this example, the dependency ratio would be 0.8.

A low dependency ratio is a significant economic advantage as there will be a large number of people in the economically active age range compared to a combination of the young dependent age group and the elderly dependent age group. In LICs, children form the great majority of the dependent population. In contrast, in HICs and MICs there is a more equal balance between young and elderly dependents.

The dependency ratio can be subdivided into the **youth dependency ratio** and the **elderly dependency ratio** as follows:

$$\text{Youth dependency ratio} = \frac{\%\text{ population aged 0–14}}{\%\text{ population aged 15–64}} (\times 100)$$

$$\text{Elderly dependency ratio} = \frac{\%\text{ population aged 65 and over}}{\%\text{ population aged 15–64}} (\times 100)$$

The dependency ratio is important because the economically active population will in general contribute more to the economy in income tax, value-added tax (VAT) and corporation tax (tax on a business). In contrast, the dependent population tend to be bigger recipients of government funding, particularly for education, healthcare and public pensions. An increase in the dependency ratio can cause significant financial problems for governments if they do not have the financial reserves to cope with such a change. Net migration can have a significant influence on the dependency ratio of a country or region. For example, a high rate of immigration of young adults and children into a HIC may:

- reduce the decline in the population under 15
- increase the number of people in the economically active population.

The **demographic dividend** is a term used by economists and others to refer to the process through which the changing age structure of a country can assist its economic growth. The general sequence leading to the demographic dividend begins with a considerable fall in fertility in a developing country that previously had a high fertility rate. This leads to a reduction in youth dependency and an increase in the proportion of the population who are of working age. In a stable economic environment, living standards should improve in terms of income per capita. Economists sometimes refer to the demographic dividend as a ‘window of opportunity’ for economic growth.

The highest demographic dividend will be in Africa, where the economically active population will increase the most. Although economic growth is due to a range of factors, the concept of the demographic dividend accounts for much of the variation in the past economic performances of different countries.

Population structure: differences within countries

In countries where there is strong rural-to-urban migration, the population structures of the areas affected can be significantly changed. These differences show up clearly on population pyramids. Out-migration from rural areas is age-selective, with single young adults and young adults with children dominating this process. Thus, population pyramids for rural areas affected by out-migration will indicate fewer people than expected in these age groups.

In contrast, population pyramids for urban areas that attract migrants will show age-selective in-migration, with substantially more young adults than expected. Such migrations may also be sex-selective. If this is the case, it should be apparent on the population pyramids.

Sex structure

The sex ratio is the number of males per 100 females in a population. The 'natural' sex ratio at birth is around 105 boys per 100 girls. Male births consistently exceed female births due to a combination of biological, social, cultural, economic and technological factors. For example:

» In many countries, more couples decide to complete their family on the birth of a boy than on the birth of a girl.
» Gender selection through sex determination is more common in some countries than others.

After birth, the gender gap generally begins to narrow until eventually females outnumber males. This is because male mortality is higher at every age than female mortality, and because women tend to live longer than men. The narrowing of the gender gap happens most rapidly in the lowest-income countries, where infant mortality is markedly higher among males than females. Here, the gap may be closed in less than a year.

In 2020, the proportion of women in the world's population was just under 50 per cent, with most countries having a female share between 49 and 51 per cent. However, there are some notable outliers:

» In South and East Asia, particularly China and India, there are significantly more males than females, due mainly to large differences in the sex ratio at birth.
» Several countries in the Middle East have a large 'excess' of males. In the UAE it is almost four to one, and in Oman it is about three to one. In both countries, the main reason is a large male migrant population.
» In many Eastern European countries there are far more females than males because of large differences in life expectancy.

Activities

1 Study Figure 6.16 (page 135). Compare the population structures of Niger and Japan, making appropriate reference to youth dependency and elderly dependency.
2 What do you understand by the term 'demographic dividend'?
3 a Define the term 'sex ratio'.
  b State two ways in which the 'natural' sex ratio can be distorted.

The issues of youthful populations

Different patterns of population growth can bring both benefits and problems to the countries concerned. This is particularly the case when countries have a very high percentage of either young people or old people in their populations.

Rapid population growth results in a large young dependent population. The global average proportion of the population under the age of 15 is 25 per cent. Table 6.8 shows the significant variation by world region around this figure.

▼ **Table 6.8** Variation by world region for the percentage of the population under the age of 15 and aged 65 and over

Region	Percentage of population under 15	Percentage of population 65 and over
World	25	10
High-income countries	16	19
Middle-income countries	25	9
Low-income countries	42	3

The 2023 World Population Data Sheet shows:

» LICs have more than two and a half times the population under 15 that HICs have.
» Twenty-nine countries had 40 per cent or more of their populations under 15. All of these countries were in Africa, with the one exception of Afghanistan.

Countries with large youthful populations have to allocate considerable resources to look after them. Young people require resources for health, education, food, water and housing above all. The money required to cover such needs may mean there is little left to invest in agriculture, industry and other aspects of the economy. A national government may see this as too large a demand on the country's resources and as a result may introduce family planning policies to reduce the birth rate. However, individual parents may have a different view, in that they see a large family as valuable in terms of the work children can do on the land. Alongside this, people in lower-income countries often have to rely on their children in old age because of the lack of state welfare benefits.

As a large young population moves up the age ladder, it will become a large working population when it enters the economically active age group. This will become an advantage if a country can attract investment and create jobs. Such a situation can create an upward spiral of economic growth. However, without investment and job creation, the unemployment rate will be high. This may cause many young adults to emigrate because of the lack of opportunities in their own country. Eventually the large number of people in this age group will reach old age, increasing elderly dependency.

The Gambia is a small LIC in Western Africa. The 2023 World Population Data Sheet gave its GNI per capita as US$ 2470. Although the country's total fertility rate fell from 6.1 in 1970 to 4.4 in 2023, the rate of natural increase is 27/1000. Forty-three per cent of the population are under 15, with only 2 per cent aged 65 years and over. The country's family planning programme has had only limited success. The Gambia's young and fast-growing population is placing big demands on its resources.

» Many parents in The Gambia struggle to provide basic housing for their families. There is huge overcrowding and a lack of sanitation, with many children sharing the same bed.
» Rates of unemployment and underemployment are high and wages are low.
» Many schools (Figure 6.17) operate a two-shift system, with one group of students attending in the morning and a different group attending in the afternoon.
» Another sign of population pressure is the large number of trees being chopped down for firewood. As a result, desertification is increasing at a rapid rate.

▲ **Figure 6.17** A school in The Gambia

The issues of demographic ageing

Ageing of population (demographic ageing) is now regarded by many experts as the most formidable population challenge facing the world. The current rate of population ageing is unprecedented in human history. Never before have so many people, in both absolute and relative terms, reached age 65 and over. Global life expectancy increased from 34 years in 1913 to 72 years in 2022, and is expected to continue on this long-term trajectory.

Demographic ageing is a rise in the median age of a population. It occurs when:

» fertility declines
» life expectancy increases
» larger cohorts of a population progress to older age groups.

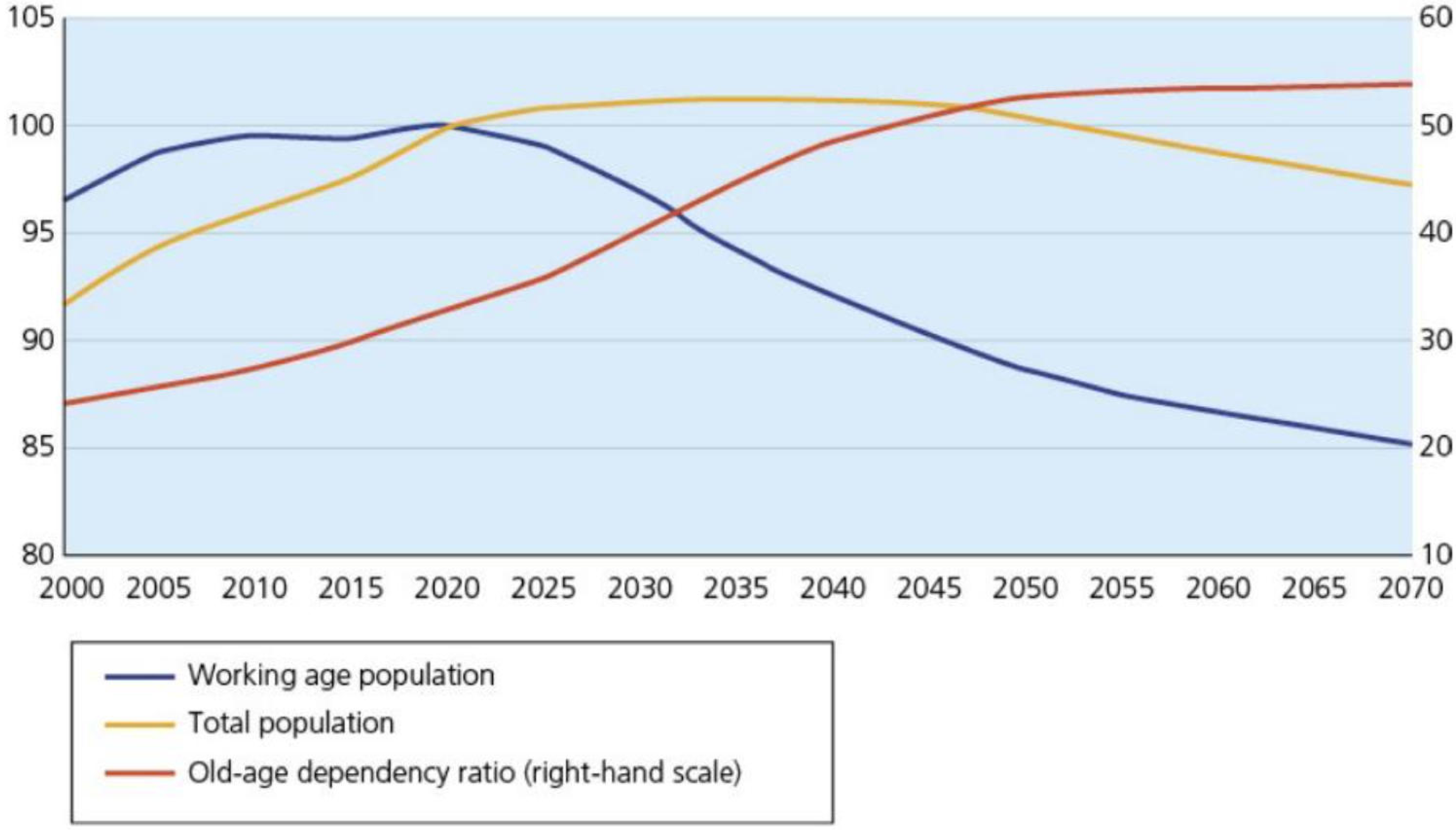
Table 6.8 (page 138) shows how the income regions of the world compare to the global average of 10 per cent of the population aged 65 years or over.

» HICs have more than six times as many of the population aged 65 and over as LICs have.
» In terms of individual countries, in 2023 Monaco topped the global league table both with a median age of 55.4 years and with 36 per cent of its population aged 65 years and over.
» For countries of a more substantial size Japan is the 'oldest', with a median age of 48.6 years and 29 per cent of its population aged 65 years and over.
» In contrast, a group of nine countries had a low of just 2 per cent of their populations aged 65 years and over in 2023 – UAE, Qatar, Chad, Burundi, Uganda, Zambia, The Gambia, Mali and Niger.

Demographic ageing puts healthcare systems and public pensions, and indeed government budgets in general, under increasing pressure. The fastest-growing segment of the population is the so called 'oldest-old': those who are 85 years old or more. It is this age group who are most likely to need expensive residential care.

Some countries have made relatively good pension provision by investing wisely over a long period of time. However, others have more or less adopted a pay-as-you-go system as the elderly dependent population rises. It is this latter group of countries that will be faced with the biggest problems in the future. At present, very few countries are generous in looking after their elderly. Poverty among the elderly is a considerable problem, but technological advances might provide a solution by improving living standards for everyone. If not, other less popular solutions, such as increased taxation, will have to be examined. Figures for the European Union (Figure 6.18) show:

» a decline in the working age population from 2020; by 2070 it will be approximately 85 per cent of the size it was in 2020
» a decline of the total population from about 2040
» an increase in the old-age dependency ratio (right-hand scale).



▲ **Figure 6.18** Working age population, total population and old-age dependency ratio for the EU, 2000–70 (left-hand scale: indices, 2020 = 100; right-hand scale: old-age dependency ratio as percentages of the 15–64 age group)

However, some demographers argue that there needs to be a certain rethinking of age and ageing, with older people adopting healthier and more adventurous lifestyles than people of the same age only a few generations ago. Sayings such as ’60 is the new 50’ have become fairly commonplace. A few years ago the United Nations established the UN Decade of Healthy Ageing (2021–30). A major theme of this initiative is ‘**demographic preparedness**’ – the idea that there is time to enact policies and encourage behaviours to limit the potential adverse effects of the demographic changes that do occur. An important behavioural change centres on increasing physical activity in all age groups, so that people are not only healthier in the earlier years of life, but also enter their advanced years in better physical and mental condition.

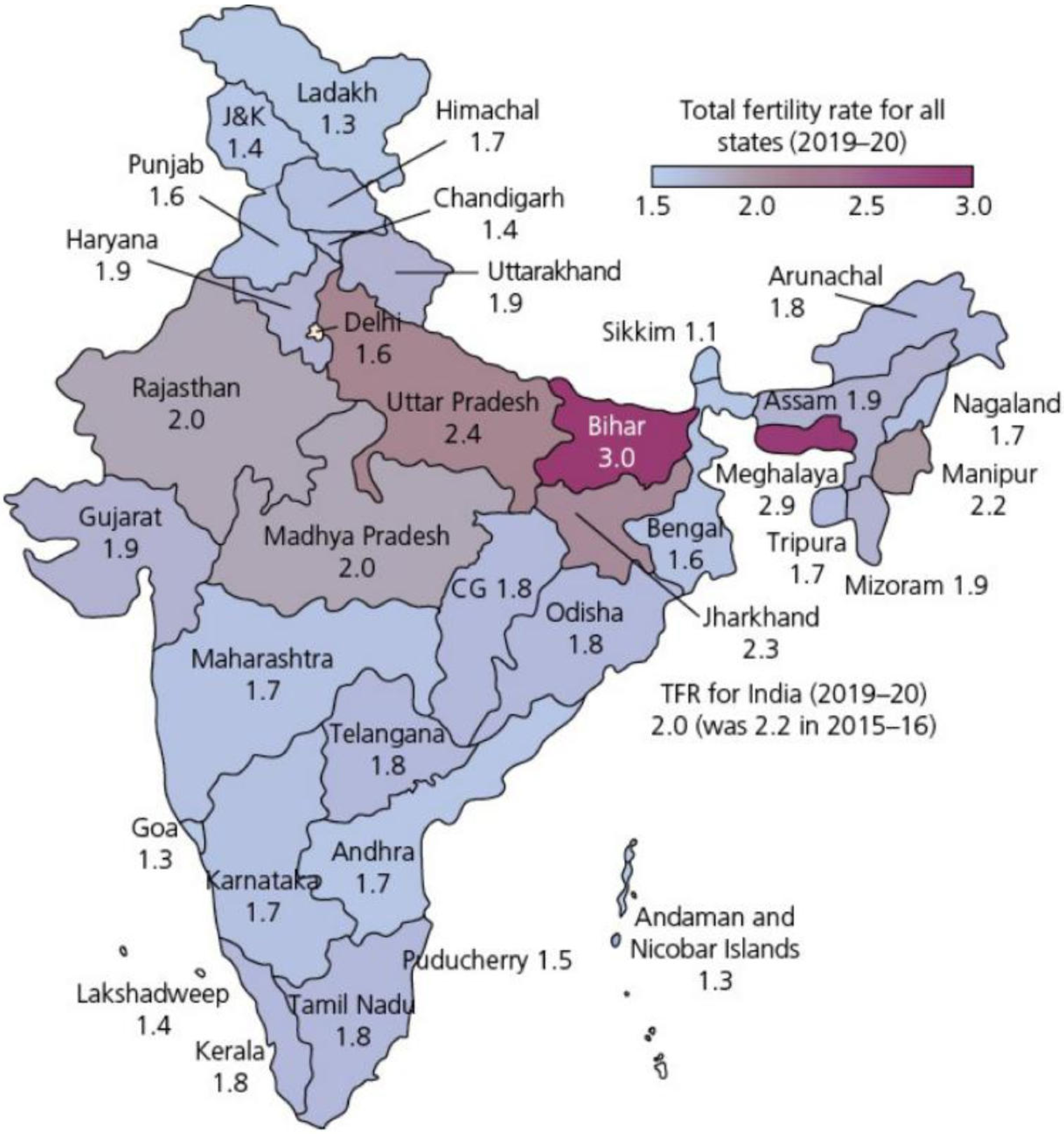
It is argued that we should think not just of chronological age, but also of prospective age – the remaining years of life expectancy people have. It is easy to underestimate the positive aspects of ageing, with many older people making a big contribution to childcare by looking after their grandchildren, while a significant number of older people work as volunteers in numerous capacities.

Detailed specific example

Renewed concerns about population size in India in the twenty-first century

In 1950, India’s population had increased to over 350 million. This was about 100 million more than the population had been in 1920, only thirty years earlier. There was a general perception in the Indian government that the country was overpopulated and that this was hindering economic development. Rural and urban birth control clinics rapidly increased in number. Two years later, India became the first country to introduce a policy designed to reduce fertility and to aid development, with a government-backed family planning programme.

- Financial and other incentives were offered in some states for those participating in programmes, especially sterilisation.
- Abortion was legalised in 1972.
- In 1978, the minimum age of marriage was increased to 18 years for females and 21 years for males.



▲ **Figure 6.19** Total fertility rate by state in India, 2019–20

In the mid-1970s the sterilisation campaign became increasingly coercive, reaching a peak of 8.3 million operations a year. International condemnation of such a coercive approach led India to return to its former approach of providing reproductive health and family planning services in an attempt to establish a stable population. Under India's federal political structure, state governments increasingly set their own priorities with regard to population policy.

India's population growth rate began to decline from 1981 and by 2020 it reached replacement level fertility, a significant point in the country's demographic transition. However, the national average figure covers wide regional differences (Figure 6.19). Table 6.9 shows the decline in India's birth rate from 1960 to 2023.

▲ **Figure 6.20** Substandard housing in a small town in northern India

▼ **Table 6.9** India's falling birth rate, 1960–2023

Year	Total population (million)	Birth rate (per 1000)
1960	451	42.5
1970	555	39.5
1980	699	36.2
1990	873	31.8
2000	1057	27.0
2010	1234	21.4
2023	1429	18.0

In very recent years, there has been a renewed high level of debate about the size of India's population. In 2019, the country's Prime Minister Narendra Modi stated that the large size of India's population was hindering its development, stating 'There is a need to have greater discussion and awareness on population explosion'. Some leading politicians have argued for the introduction of a national two-child policy, but opponents view such a policy as a return to coercion.

India now has the world's largest population at 1.43 billion (2023). Under the UN's 'medium variant' projection, its population will surpass 1.5 billion by the end of the 2020s. India's population will continue to grow slowly until it peaks at 1.7 billion in 2064. The country has a working age population of about 500 million and it is still expanding. Some observers see this situation as a demographic dividend, but others see it as potentially an unsustainable drain on the country's resources. Greater investment in education and training will be vital for a positive outcome. Despite declining fertility, population momentum will see the total population increase for another four decades before it stabilises. The median age in India is 28 years, compared to 39 years in China.

An important body of opinion stresses the need to focus on the high-fertility regions of the country (Figure 6.19). The Indian government has identified 146 high-fertility districts. Most are in the northern states of Bihar, Rajasthan, Jharkhand, Madhya Pradesh and Uttar Pradesh. All these states have well below average development scores in education and health. New initiatives have focused on increasing supplies of contraceptives and on family planning campaigns.

Bihar, with a 2021 population of 128 million, is the most densely populated state in India. The state's population grew by over 18 per cent between 2011 and 2020, the highest growth in India. The fertility rate has resulted in a high young dependent population. Bihar has lagged behind India in general in socioeconomic development. A considerable proportion of the population live below the poverty line.

In March 2023, the Indian government stated that the minimum age of marriage for women would increase from 18 years of age to 21, putting it on a par with the age for men. At the time of writing, the proposal was progressing through parliament. The new legislation will come into effect two years after it is agreed. A major Indian media organisation has termed the forthcoming policy 'a momentous decision'.

Southern states such as Kerala and Tamil Nadu have adopted an approach to fertility more related to overall development. As levels of education, health and GDP have improved, fertility rates have declined significantly. Some academics have stressed the importance of India utilising its **gender dividend**. This is the increase in economic growth that can result from greater investment in women and girls, including promoting secondary and tertiary education and female workforce participation. According to the World Bank, just 23 per cent of Indian women perform paid work, compared to 63 per cent in China.

Activities

1 Study Table 6.9. Discuss the reasons behind the steady decline of India's birth rate.
2 Why have some academics stressed the importance of India utilising its gender dividend?